Job Portal Management System - MySQL Capstone Project

Project Objective: Design and implement a Job Portal database using MySQL to manage candidates, employers, jobs, applications and interviews.

Database Design

Candidates(candidate_id PK, full_name, phone UNIQUE, email UNIQUE, city)
Employers(employer_id PK, company_name, industry)
Jobs(job_id PK, employer_id FK, title, location, salary)
Applications(application_id PK, candidate_id FK, job_id FK, application_date, status)
Interviews(interview_id PK, application_id FK, interview_date, mode, result)

Sample Data

```
-- Candidates
INSERT INTO Candidates VALUES
(1,'Aarav','900000001','c1@mail.com','Hyderabad'),
(2,'Bhavya','900000002','c2@mail.com','Vijayawada'),
(3,'Charan','900000003','c3@mail.com','Vizag'),
(4,'Divya','900000004','c4@mail.com','Guntur'),
(5,'Esha','900000005','c5@mail.com','Warangal'),
(6,'Farhan','900000006','c6@mail.com','Nellore'),
(7,'Gopi','900000007','c7@mail.com','Kurnool'),
(8,'Hari','900000008','c8@mail.com','Tirupati'),
(9,'Isha','900000009','c9@mail.com','Chennai'),
(10,'John','900000010','c10@mail.com','Bengaluru');

-- Employers
INSERT INTO Employers VALUES
(101,'TechNova','IT'),(102,'DataWave','Analytics'),(103,'CloudEdge','Cloud'),
(104,'FinCore','Finance'),(105,'HealthSoft','Healthcare');

-- Jobs
INSERT INTO Jobs VALUES
(1001,101,'Python Developer','Hyderabad',650000),
(1002,101,'SQL Developer','Hyderabad',550000),
(1003,102,'Data Analyst','Vizag',600000),
(1004,103,'Cloud Engineer','Bengaluru',850000),
(1005,104,'BI Developer','Chennai',700000),
(1006,105,'ML Engineer','Hyderabad',950000);
```

```
-- Applications
INSERT INTO Applications VALUES
(1,1,1001,'2026-07-01','Applied'),
(2,2,1003,'2026-07-02','Shortlisted'),
(3,3,1002,'2026-07-02','Rejected'),
(4,4,1004,'2026-07-03','Interview'),
(5,5,1005,'2026-07-03','Selected'),
(6,6,1006,'2026-07-04','Applied'),
(7,7,1001,'2026-07-05','Interview'),
(8,8,1002,'2026-07-06','Shortlisted'),
(9,9,1004,'2026-07-07','Applied'),
(10,10,1006,'2026-07-08','Selected');

-- Interviews
INSERT INTO Interviews VALUES
(201,2,'2026-07-05','Online','Pending'),
(202,4,'2026-07-06','Offline','Pending'),
(203,5,'2026-07-07','Online','Selected'),
(204,7,'2026-07-08','Offline','Rejected'),
(205,8,'2026-07-09','Online','Pending'),
(206,10,'2026-07-10','Offline','Selected');
```

40 Business Queries

1. List all jobs.

3. Show applications with candidate names.

4. Display employer names in uppercase/lowercase.

5. Show the first five characters of every company name.

6. Concatenate the candidate's name and city.

7. Replace the word **Engineer** with **Specialist** in job titles.

8. Find all jobs where the title contains the word **Developer**.

9. Find jobs in Hyderabad.

10. Applications in last 7 days.

11. Count jobs company-wise.

12. Format application dates in the format: Mon-Jul-26

13. Find candidates whose interviews are scheduled today.

14. Show the number of days since each application was submitted.

15. Display the highest and lowest salary offered.

16. Calculate the total salary budget offered by each company.

17. Count applications by status.

18. Employer with most jobs.

19. Candidate with most applications.

20. Selected candidates.

21. Rejected applications.

22. Companies with more than one job.

23. Show locations having more than two job openings.

24. Average salary by company.

25. Top 5 latest jobs.

26. Jobs in IT industry.

27. Display candidate name, company name, and job title for every application.

28. Display interview details along with candidate and company information.

29. Display company names with the total number of applications received.

30. Candidates without applications.

31. Display candidates who have never attended an interview.

32. Create a view named **Selected_Candidates** showing selected candidates with company and job details.

33. Create a view to display all active job postings.

34. Retrieve data from the created views.

35. Start a transaction and update application status to Interview Scheduled.

36. Roll back a transaction after deleting an application accidentally.

37. Create a stored procedure to display all jobs offered by a particular employer.

- Input: employer_id
- Display job title, location, and salary.

38. Create a stored procedure to display all applications with a particular status.

- Input: status
- For example, display all Shortlisted applications.

39. Create a trigger on the Applications table.

- Whenever a new application is inserted, automatically store the application_id, candidate_id, job_id, and application date in an Application_Log table.

40. Create a trigger on the Interviews table.

- Whenever an interview record is deleted, automatically store the deleted interview details in an Interview_Archive table.